

Functional Annotation of *xpt* — Xanthine Phosphoribosyltransferase (Q88CB6, PP_5265)

Pseudomonas putida KT2440

0. Gene/Protein Identity Verification

Before research, the target identity was confirmed against UniProt and matches the literature used here:

Attribute	Value (verified)
UniProt accession	Q88CB6 (reviewed / Swiss-Prot)
Gene name	xpt ; ordered locus PP_5265
Protein	Xanthine phosphoribosyltransferase (XPRTase), EC 2.4.2.22
Length	190 aa; homodimer ; cytoplasm
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / KT2440)
Family	Purine/pyrimidine phosphoribosyltransferase, Xpt subfamily (HAMAP MF_01184)
Domains	Type I PRTase (PRTase-like, IPR029057; PRTase_dom, IPR000836; Xanthine_PRibTrfase, IPR010079)

The gene symbol *xpt* is **unambiguous and correct** for this protein: it is the canonical symbol for xanthine phosphoribosyltransferase, and the domain architecture, EC number, family, and reaction all align with the literature cited below. (Note: the *E. coli* gene *gpt*, encoding xanthine-**guanine** PRT / XGPRT, and human/parasite *hprt/hpt* are related but distinct family members

with broader base specificity; where their structural/mechanistic data are used below, this is explicitly to inform the conserved catalytic machinery of the type I PRTase family, not to substitute a different gene.)

1. Summary (Answer to the Research Question)

xpt encodes **xanthine phosphoribosyltransferase (XPRTase, EC 2.4.2.22)**, a **cytoplasmic, Mg^{2+} -dependent enzyme of the purine *salvage* pathway**. Its primary and essentially sole catalytic function is to transfer the 5-phosphoribosyl group of 5-phospho- α -D-ribose-1-diphosphate (PRPP) onto the N9 nitrogen of the free purine base **xanthine**, producing **xanthosine-5'-monophosphate (XMP)** and inorganic pyrophosphate. This single reaction ("XMP from xanthine: step 1/1") reclaims xanthine — a product of nucleic-acid and purine turnover — into the nucleotide pool, where XMP is converted to GMP by GMP synthetase and channels into guanine-nucleotide metabolism. The enzyme acts in the **cytosol**, works as a homodimer of a type I PRTase fold, and is genomically and regulatorily embedded in a xanthine-utilization module together with a xanthine permease (pbuX-type), controlled by purine availability.

2. Primary Molecular Function — the Reaction Catalyzed

Reaction (UniProt Q88CB6 / HAMAP MF_01184):



i.e. in the physiological (salvage) direction:



UniProt describes the function as: "Converts the preformed base xanthine, a product of nucleic acid breakdown, to xanthosine 5'-monophosphate (XMP), so it can be reused for RNA or DNA synthesis." The metabolic pathway assignment is "Purine metabolism; XMP biosynthesis via salvage pathway; XMP from xanthine: step 1/1" — meaning xpt performs the complete, one-step salvage conversion of xanthine to a nucleotide.

Substrate specificity. The Xpt subfamily is defined by *specific* recognition of xanthine, in contrast to the broader hypoxanthine-guanine PRTs (HGPRT) or the xanthine-guanine PRT (XGPRT/*gpt*) of *E. coli*. This specificity was established biochemically for the founding member of the subfamily in *Bacillus subtilis*, where "**the xpt gene encodes a specific xanthine phosphoribosyltransferase**" [Christiansen et al., 1997,

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The *P. putida* protein Q88CB6 is a direct orthologue in the same Xpt subfamily and is annotated with the identical dedicated activity.

Direction / thermodynamics. Like all type I purine PRTases the reaction is formally reversible (pyrophosphorolysis is measurable in vitro), but in vivo it is driven forward by (i) the high group-transfer potential of PRPP and (ii) hydrolysis of the released pyrophosphate by inorganic pyrophosphatase, making salvage effectively unidirectional under cellular conditions.

3. Catalytic Mechanism and Structural Basis

XPRTase belongs to the **type I PRTase structural family**, built on a Rossmann-like α/β core domain that binds PRPP plus a smaller "hood" subdomain that caps the purine base (IPR029057/IPR000836/IPR010079). Precise mechanistic understanding derives from high-resolution structures and kinetics of the closely related 6-oxopurine PRTases (HPRT/HGPRT/HG(X)PRT), whose catalytic residues and reaction chemistry are conserved across the family:

- 1. Ordered sequential (bi-bi) kinetics.** Substrates add in a defined order — **PRPP (with Mg^{2+}) binds first, then the purine base** — and products leave in order — **pyrophosphate first, then the nucleotide**. This was shown by steady-state kinetics of *Tritrichomonas foetus* HG(X)PRTase: "*the substrates bind to the enzyme (first PRPP followed by the purine bases), and the products released (first PPi followed by purine nucleotide) in a defined order*" [Munagala et al., 1998,

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2. **Two-metal-ion ribosyl transfer.** The *Trypanosoma cruzi* HPRT ternary complex revealed that *"the ligands are positioned for in-line nucleophilic attack at the PRPP ribose C1' by two metal ions which straddle the pyrophosphate leaving group,"* consistent with SN2-type/oxocarbenium-stabilized chemistry at the anomeric carbon [Focia et al., 1998,

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Catalysis inverts the anomeric configuration, forming the β -N9-glycosidic bond of the nucleotide.

3. **Flexible catalytic loop closure.** A mobile active-site loop closes over the bound substrates to sequester the reaction from solvent and to position the pyrophosphate for departure. Free vs. liganded human HGPRT structures show that *"significant conformational changes have to occur for the substrate(s) to bind and for catalysis to proceed"* [Keough et al., 2005,

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the conserved Ser-Tyr loop dipeptide participates in PP_i liberation.

4. **Cofactor and quaternary structure.** The enzyme requires Mg^{2+} (as the Mg-PRPP complex and as active-site catalytic metal) and functions as a **homodimer** in *P. putida* (UniProt Q88CB6); other family members are dimers or tetramers.

These features are directly transferable to Q88CB6 by strong sequence/structural homology within the Xpt subfamily, and constitute *inference from structure and evolution* for the *P. putida* enzyme.

3.5 Active-site residues mapped on Q88CB6 itself

Curated UniProt features (HAMAP MF_01184) place the catalytic machinery on specific residues of the 190-aa *P. putida* chain, providing direct sequence-level evidence rather than pure homology transfer:

Ligand	Residues in Q88CB6	Note
Xanthine (purine base)	Leu20, Asn27, Lys156	base recognition in the "hood" region; specificity determinants
PRPP (5-phosphoribosyl-1-PP)	loop 128–132 (...ANGK A...)	phosphate/ribose binding
Twin-aspartate motif	Asp-Asp within "...V I D D F L A N G K..." (~res 127–129)	invariant type I PRTase PRPP-binding signature (PROSITE PS00103); coordinates ribose-5-phosphate and catalytic Mg ²⁺

The presence and correct spacing of the diagnostic Asp-Asp PRPP loop plus dedicated xanthine contacts confirm Q88CB6 is a catalytically competent, xanthine-specific type I PRTase independent of homology arguments alone.

3.6 Structural-model confidence (AlphaFold)

The AlphaFold DB model of Q88CB6 (v6, 190 aa) is of very high quality — **mean pLDDT 94.9** (median 97.8; 88 % of residues > 90, 97 % > 70, none < 50) — describing a single, well-ordered globular domain with no substantial disordered regions. Critically, all curated ligand-binding residues are modelled at very-high confidence (Leu20 = 97.4, Asn27 = 97.3, Lys156 = 97.6; PRPP loop 128–132 = 94–95), giving independent structural support that the enzyme adopts a properly constituted type I PRTase fold (Rossmann-like PRPP core + purine "hood") with an intact xanthine/PRPP active site.

4. Subcellular Localization

XPRTase is a soluble **cytoplasmic** enzyme (UniProt Q88CB6, subcellular location: Cytoplasm), consistent with (i) its role in intracellular nucleotide metabolism, (ii) its use of the cytosolic metabolites PRPP and free purine bases, and (iii) the absence of signal peptides or membrane-spanning segments in the 190-residue chain. The upstream/co-operonic xanthine permease (PbuX-type) is the membrane component that supplies the base; the PRTase itself acts in the cytosol immediately downstream of import.

5. Pathway Context and Biological Role

5.1 Position in purine salvage

xpt is the committed, terminal step that converts free xanthine into a nucleotide. Its product **XMP is the direct precursor of GMP**: $\text{XMP} \rightarrow \text{GMP}$ is catalyzed by GMP synthetase (guaA). Thus xanthine salvaged by XPRTase feeds specifically into the **guanine nucleotide pool** ($\text{XMP} \rightarrow \text{GMP} \rightarrow \text{GDP/GTP, dGTP}$). This wiring is made explicit by the conserved operon in *Staphylococcus aureus*, "an operon encoding xanthine phosphoribosyltransferase (xpt), xanthine permease (pbuX), inosine-5'-monophosphate dehydrogenase (guaB), and GMP synthetase (guaA)" [Kofoed et al., 2016,

P 27161118

— grouping the transporter, the salvage PRTase, and the two enzymes that convert its product toward GMP.

5.2 The xanthine-utilization module (transport + salvage) — and its divergence in *P. putida*

xpt is functionally linked to a **xanthine-specific permease** that supplies its substrate. In *B. subtilis*, xpt and pbuX have overlapping reading frames and "an operon organization," with pbuX translation dependent on xpt translation [PMID 9098051]; the same xpt-pbuX coupling recurs in *S. aureus* [PMID 27161118]. The logic is: import xanthine, then immediately phosphoribosylate it.

Important organism-specific correction (this work). This operonic coupling is **not** conserved in *P. putida* KT2440. Genome-context analysis (KEGG) shows **xpt = PP_5265 is monocistronic**: it lies at 6,013,223–6,013,795 flanked by functionally unrelated genes — a Rep-type ssDNA-dependent helicase (PP_5264) and an acetyl-CoA-hydrolase-family protein (PP_5266), with a GGDEF protein (PP_5263) and a putative cytochrome c5 (PP_5267) nearby. **No xanthine permease is adjacent.** The candidate xanthine/uracil (NCS2-family) permeases of KT2440 (PP_3655, PP_4035, PP_4309) are dispersed elsewhere on the chromosome and are genomically unlinked to xpt. Thus in *P. putida* the salvage *reaction* is conserved, but xanthine transport is supplied **in trans** by separately encoded permeases rather than by an operon-linked pbuX. Consequently the riboswitch-in-operon regulatory model established in Firmicutes (§5.3) should **not** be assumed to hold in this organism.

5.3 Regulation — responsive to purine availability

Because salvage is only useful when preformed purines are present, xpt expression is tuned to purine levels: - **Purine repression / riboswitch control.** The *B. subtilis* *xpt-pbuX* operon is preceded by a **guanine riboswitch** that "directly binds guanine, hypoxanthine or xanthine to terminate transcription" [Batey et al., 2004,

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the same aptamer structural class characterized in].

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Expression is repressed up to 160-fold by hypoxanthine+guanine via transcription antitermination [PMID 9098051]. This couples XPRTase production to intracellular purine sufficiency. - **Physiological dispensability.** As a salvage (not de novo) function, xpt is individually non-essential when de novo synthesis operates: in *S. aureus*, "deletion of the purine salvage genes *xpt-pbuX* had none of these effects" on growth/virulence, whereas loss of the downstream de novo/GMP enzymes was deleterious [PMID 27161118]. This pinpoints xpt's role as an economizing salvage route rather than an indispensable biosynthetic step.

5.4 Organism-specific relevance in *P. putida*

P. putida is a metabolically versatile soil bacterium capable of purine and methylxanthine catabolism. Classic work on *P. putida* showed degradation of xanthine through **uric acid → allantoin → allantoic acid** (e.g., during caffeine/methylxanthine breakdown) [Blecher & Lingens, 1977,

P 561017

XPRTase sits at the **branch point** between this catabolic fate of xanthine (oxidation to urate for use as C/N source) and the **anabolic salvage** fate (phosphoribosylation to XMP for nucleotide synthesis). By capturing xanthine as XMP, xpt allows the cell to recycle purine bases into GTP/dGTP pools rather than committing them to oxidative catabolism, thereby conserving carbon, nitrogen, and biosynthetic energy.

6. Evidence Summary

Claim	Type of evidence	Source
xpt = xanthine PRTase, EC 2.4.2.22; reaction xanthine+PRPP→XMP+PP _i ; salvage step 1/1; cytoplasm; homodimer	Curated database annotation (HAMAP rule, reviewed)	UniProt Q88CB6
Xpt is <i>specific</i> for xanthine (defines subfamily specificity)	Direct biochemical (cloning + enzyme characterization)	P 9098051
Two-metal-ion, in-line ribosyl transfer at PRPP C1'	Experimental structure (ternary complex) of homologue	P 9860824
Ordered bi-bi kinetics (PRPP first; PP _i released first)	Steady-state kinetics of homologue	P 9521725
Flexible catalytic-loop conformational change during catalysis	Experimental structures (apo vs. liganded)	P 15990111
xpt-pbuX operon (transport+salvage coupling)	Genetic/transcriptional (operon mapping)	P 9098051
xpt→XMP feeds guaB/guaA→GMP; salvage genes dispensable	Genetic (operon architecture, deletions)	P 27161118
Purine/guanine-riboswitch regulation of xpt operon	Structural + genetic	P 15549109 ; 9098051; 17959930
Xanthine catabolism (urate/allantoin) in <i>P. putida</i>	Biochemical pathway characterization	P 561017
Q88CB6 active site: xanthine at L20/N27/K156; PRPP loop 128–132 with twin-Asp motif	Curated sequence features / motif analysis	UniProt Q88CB6; PROSITE PS00103
xpt (PP_5265) is monocistronic in <i>P. putida</i> ; NCS2 permeases (PP_3655/4035/4309) unlinked	Genome-context (bioinformatic) analysis	KEGG ppu:PP_5265 & neighbours
Q88CB6 folds as a confident type I PRTase (mean pLDDT 94.9; active-site residues >94)	Structural prediction	AlphaFold DB Q88CB6 (v6)

Evidence tiers: Q88CB6-specific claims (reaction, EC, pathway, localization, oligomeric state) rest on curated, rule-based database annotation. Substrate specificity is supported by direct biochemistry on the orthologous *B. subtilis* Xpt. Mechanistic details are strong *inferences from structure and evolution* using experimentally solved structures/kinetics of closely related type I purine PRTases (HPRT/HG(X)PRT), whose catalytic apparatus is conserved.

7. Supported and Refuted Hypotheses

- **Supported:** *xpt* catalyzes xanthine + PRPP → XMP + PP_i (dedicated xanthine salvage). ✓
 - **Supported:** The enzyme is cytoplasmic, Mg²⁺-dependent, homodimeric, type I PRTase fold with ordered bi-bi mechanism. ✓ (annotation + homology/mechanistic inference)
 - **Supported:** XMP product feeds guanine-nucleotide synthesis (via *guaB/guaA*→GMP); *xpt* is a purine-availability-regulated salvage function. ✓
 - **Refuted / not supported:** That *P. putida* *xpt* is a broad-specificity xanthine-**guanine** PRT like *E. coli* *gpt* — the Xpt subfamily and the founding biochemistry indicate *xanthine-specific* activity, not the dual xanthine/guanine XGPRT. ✗
 - **Refuted / not supported:** That *xpt* is an essential/de novo biosynthetic enzyme — it is a dispensable salvage route under de novo-competent conditions. ✗
 - **Refuted for *P. putida* (revised in Iteration 2):** That *xpt* is operon-coupled to a *pbuX* xanthine permease in *P. putida* as in Firmicutes — genome context shows PP_5265 is monocistronic; transport is supplied in trans by unlinked NCS2 permeases. ✗
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8. Limitations and Future Directions

1. **No direct enzymology on Q88CB6 itself.** Kinetic constants (K_m for xanthine and PRPP, k_{cat}), metal-ion stoichiometry, and an experimental structure have not, to my knowledge, been reported specifically for the *P. putida* KT2440 protein; the functional assignment relies on curated annotation and orthology to the biochemically characterized *B. subtilis* Xpt and to structurally characterized family members. Recombinant expression + steady-state kinetics and a crystal/cryo-EM or AlphaFold-validated structure would confirm specificity and mechanism directly.

2. **Regulatory architecture in *P. putida* not fully mapped.** The guanine-riboswitch/purine-repression model is best established in Gram-positive Firmicutes; the precise *cis*-regulatory elements and operon boundaries around PP_5265 in *P. putida* KT2440 should be verified transcriptomically.
 3. **Metabolic partitioning.** The quantitative flux split between xanthine salvage (via Xpt) and xanthine catabolism (urate/allantoin) in *P. putida* under different C/N regimes is an open, experimentally testable question (e.g., $^{13}\text{C}/^{15}\text{N}$ tracing, Δxpt mutant phenotyping on xanthine as N source).
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Prepared as an autonomous functional-annotation report. Finalized in Iteration 3 of 3. Six findings recorded across iterations: reaction/specificity, mechanism, salvage-module context, Q88CB6 active-site mapping, P. putida monocistronic genomic context, and AlphaFold structural confidence.

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